

Differentiation

- Basic Rules of Differentiation
- The Product and Quotient Rules
- The Chain Rule
- Marginal Functions in Economics
- Higher-Order Derivatives
- Implicit Differentiation and Related Rates
- Differentials

Basic Differentiation Rules

1. $\frac{d}{dx}(c) = 0 \quad (c \text{ is a constant})$

Ex. $f(x) = 5$
 $f'(x) = 0$

2. $\frac{d}{dx}(x^n) = nx^{n-1} \quad (n \text{ is a real number})$

Ex. $f(x) = x^7$
 $f'(x) = 7x^6$

Basic Differentiation Rules

$$3. \frac{d}{dx}(cf(x)) = c \frac{d}{dx}(f(x)) \quad (c \text{ is a constant})$$

Ex. $f(x) = 3x^8$

$$f'(x) = 3(8x^7) = 24x^7$$

$$4. \frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$$

Ex. $f(x) = 7 + x^{12}$

$$f'(x) = 0 + 12x^{11} = 12x^{11}$$

More Differentiation Rules

5. Product Rule

$$\frac{d}{dx}[f(x) \cdot g(x)] = \frac{d}{dx}[f(x)]g(x) + \frac{d}{dx}[g(x)]f(x)$$

Ex. $f(x) = (x^3 + 2x + 5)(3x^7 - 8x^2 + 1)$

$$f'(x) = (3x^2 + 2)(3x^7 - 8x^2 + 1) + (x^3 + 2x + 5)(21x^6 - 16x)$$

Derivative
of the first
function

Derivative of
the second
function

$$f'(x) = 30x^9 + 48x^7 + 105x^6 - 40x^4 - 45x^2 - 80x + 2$$

More Differentiation Rules

6. Quotient Rule

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} [f(x)] - f(x) \frac{d}{dx} [g(x)]}{[g(x)]^2}$$

More Differentiation Rules

6. Quotient Rule (cont.)

Ex. $f(x) = \frac{3x + 5}{x^2 - 2}$

Derivative of
the numerator



Derivative of
the denominator



$$\begin{aligned} f'(x) &= \frac{3(x^2 - 2) - 2x(3x + 5)}{(x^2 - 2)^2} \\ &= \frac{-3x^2 - 10x - 6}{(x^2 - 2)^2} \end{aligned}$$

More Differentiation Rules

7. The Chain Rule

If $h(x) = g(f(x))$ then

$$h'(x) = g'(f(x)) \cdot f'(x)$$

Note: $h(x)$ is a composite function.

Another Version:

If $y = h(x) = g(u)$, where $u = f(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

More Differentiation Rules

The Chain Rule leads to

The General Power Rule:

If $h(x) = [f(x)]^n$ (n , real) then

$$h'(x) = n[f(x)]^{n-1} \cdot f'(x)$$

Ex. $f(x) = \sqrt{3x^2 + 4x} = (3x^2 + 4x)^{1/2}$

$$\begin{aligned} f'(x) &= \frac{1}{2} (3x^2 + 4x)^{-1/2} (6x + 4) \\ &= \frac{3x + 2}{\sqrt{3x^2 + 4x}} \end{aligned}$$

Chain Rule Example

Ex. $G(x) = \left(\frac{2x-1}{3x+5} \right)^7$

$$G'(x) = 7 \left(\frac{2x-1}{3x+5} \right)^6 \left(\frac{(3x+5)2 - (2x-1)3}{(3x+5)^2} \right)$$

$$G'(x) = 7 \left(\frac{2x-1}{3x+5} \right)^6 \frac{13}{(3x+5)^2} = \frac{91(2x-1)^6}{(3x+5)^8}$$

Chain Rule Example

Ex. $y = u^{5/2}, u = 7x^8 + 3x^2$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= \frac{5}{2} u^{3/2} \cdot (56x^7 + 6x)$$

Sub in for u



$$= \frac{5}{2} (7x^8 + 3x^2)^{3/2} \cdot (56x^7 + 6x)$$

$$= (140x^7 + 15x) (7x^8 + 3x^2)^{3/2}$$

Higher Derivatives

The second derivative of a function f is the derivative of the derivative of f at a point x in the domain of the first derivative.

Derivative	Notations	
Second	f''	$\frac{d^2 y}{dx^2}$
Third	f'''	$\frac{d^3 y}{dx^3}$
Fourth	$f^{(4)}$	$\frac{d^4 y}{dx^4}$
n th	$f^{(n)}$	$\frac{d^n y}{dx^n}$

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Example of Higher Derivatives

Given $f(x) = 3x^5 - 2x^3 + 14$ find $f'''(x)$.

$$f'(x) = 15x^4 - 6x^2$$

$$f''(x) = 60x^3 - 12x$$

$$f'''(x) = 180x^2 - 12$$

Example of Higher Derivatives

Given $f(x) = \frac{2x+1}{3x-2}$ find $f''(2)$.

$$f'(x) = \frac{2(3x-2) - 3(2x+1)}{(3x-2)^2} = \frac{-7}{(3x-2)^2} = -7(3x-2)^{-2}$$

$$f''(x) = 14(3x-2)^{-3} (3) = \frac{42}{(3x-2)^3}$$

$$f''(2) = \frac{42}{(3(2)-2)^3} = \frac{42}{4^3} = \frac{21}{32}$$

Implicit Differentiation

$$y = 3x^3 - 4x + 17$$

 y is expressed *explicitly* as a function of x .

$$y^3 + xy = 3x + 1$$

 y is expressed *implicitly* as a function of x .

To differentiate the implicit equation, we write $f(x)$ in place of y to get:

$$[f(x)]^3 + x[f(x)] = 3x + 1$$

Implicit Differentiation (cont.)

Now differentiate $[f(x)]^3 + x[f(x)] = 3x + 1$

using the chain rule:

$$3[f(x)]^2 f'(x) + f(x) + xf'(x) = 3$$

which can be written in the form

$$3y^2 y' + y + xy' = 3$$

subbing in y

$$y'(3y^2 + x) = 3 - y$$

Solve for y'

$$y' = \frac{3 - y}{3y^2 + x}$$

Related Rates

Look at how the rate of change of one quantity is related to the rate of change of another quantity.

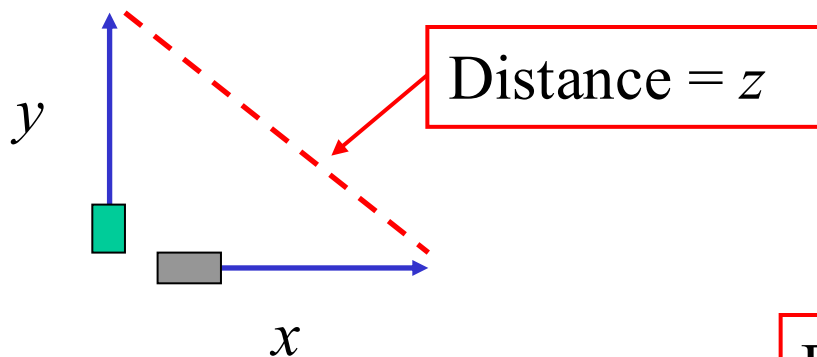
Ex. Two cars leave an intersection at the same time. One car travels north at 35 mi./hr., the other travels west at 60 mi./hr. How fast is the distance between them changing after 2 hours?

Note: The rate of change of the distance between them is *related* to the rate at which the cars are traveling.

Ex. Two cars leave an intersection at the same time. One car travels north at 35 mi./hr., the other travels east at 60 mi./hr. How fast is the distance between them changing after 2 hours?

$$\frac{dx}{dt} = 60 \quad \frac{dy}{dt} = 35$$

$$x = 120 \quad y = 70$$



$$x^2 + y^2 = z^2$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2z \frac{dz}{dt}$$

$$2(120)(60) + 2(70)(35) = 2(10\sqrt{193}) \frac{dz}{dt}$$

$$\frac{dz}{dt} \approx 69.5 \text{ mi./hr.}$$

From original
relationship

Increments

An increment in x represents a change from x_1 to x_2 and is defined by:

$$\Delta x = x_2 - x_1$$

Read “delta x ”



An increment in y represents a change in y and is defined by:

$$\Delta y = f(x + \Delta x) - f(x)$$

Differentials

Let $y = f(x)$ be a differentiable function, then the differential of x , denoted dx , is such that $dx = \Delta x$. The differential of y , denoted dy , is

$$dy = f'(x)\Delta x = f'(x)dx$$

Note: Δy measures actual change in y
 dy measures approximate change in y

Example

Given $f(x) = 3x^2 - x$, find:

1. Δx as x changes from 3 to 3.02.

$$\Delta x = 3.02 - 3 = 0.02$$

2. Δy and dy as x changes from 3 to 3.02.

$$\begin{aligned}\Delta y &= f(3.02) - f(3) \\ &= 24.3412 - 24 = 0.3412\end{aligned}$$

$$\begin{aligned}dy &= f'(x)dx = (6x - 1)dx \\ &= (6(3) - 1)(0.02) = 0.34\end{aligned}$$